MINI-PROJECT SHORT TECHNICAL REPORT

Course: Cross-Platform Mobile App Development (VKU)

Mini-Project Title: Mini-Project 3: OCR Expense Tracker & Receipt Parser (Flutter 3.x & Dart 3)

Team / Student Name: Nguyễn Thị Thương

Submission Date: 12/10/2026

1. GENERAL INFORMATION & DELIVERABLE LINKS

* **Team Members:**

1. **Nguyễn Thị Thương** — Student ID: **23IT.B219** — Class: **23SE4** — Course Class: **Phát triển ứng dụng di động đa nền tảng (4)** — Instructor: **TS. Nguyễn Thanh Tuấn** — Role: **Full-Stack Mobile & On-Device AI Developer** — Contribution: **100%**

* 🔗 **Live Demo URL:** https://vku-ocr-expense.thuong2213.workers.dev/

* 📱 **Signed Release APK (`app-release.apk`):** https://github.com/thuong614oh65/vku-ocr-expense-tracker/releases/download/v1.0.0/app-release.apk

* 💻 **GitHub Repository:** https://github.com/thuong614oh65/vku-ocr-expense-tracker

2. FEATURE IMPLEMENTATION CHECKLIST (10-POINT RUBRIC ALIGNMENT)

#	Required Feature	Status	Implementation Details & Acceptance Level
1	**Camera Capture, Focus Tap & Framing Crop**	✅ Complete	`CameraScannerScreen` + `CameraOverlayPainter`: Live camera viewfinder with Flash/Torch toggle, tap-to-focus (`AF`) reticle animation, and even-odd framing crop overlay with animated laser.
2	**On-Device Text Recognition (ML Kit)**	✅ Complete	`OcrService` integrates `google_mlkit_text_recognition` (`TextRecognizer`) for offline, sub-100ms on-device text extraction with zero cloud cost.
3	**Regex Heuristic Parser & Dart 3 Records**	✅ Complete	`RegexReceiptParser` uses multi-stage regex heuristics & Dart 3 Records `(String merchant, double total, DateTime date)` to extract Vietnamese totals (`150.000 đ`, `150,000 VND`), dates (`DD/MM/YYYY`), merchants, and 6 categories.
4	**Interactive Review & Form Validation**	✅ Complete	`ReceiptReviewScreen` uses `Form`, `GlobalKey`, `AutovalidateMode.onUserInteraction`, `TextFormField` validators, `FocusNode` chaining, AI confidence badge, and raw OCR inspector.
5	**Riverpod 2 Compile-Safe State Management**	✅ Complete	`expense_provider.dart` implements `ExpenseController extends Notifier<>`, `ThemeModeController`, and `CategoryFilterController` consumed via `ConsumerStatefulWidget`.
6	**Local SQLite (`sqflite`) & Thumbnail Cache**	✅ Complete	`StorageService` persists transactions in `sqflite` (`vku_expense_ocr.db`) with Web `SharedPreferences` fallback, and caches cropped receipt thumbnails in the app documents directory (`path_provider`).
7	**CustomPainter Animated Donut / Pie Chart**	✅ Complete	`CustomPieChart` draws category distribution directly on GPU Canvas via `canvas.drawArc` + `AnimationController` (`Curves.easeOutCubic`) without any third-party chart library.
8	**CustomPainter Animated Weekly Bar Chart**	✅ Complete	`CustomBarChart` renders 7-day (`T2 → CN`) spending bars using `canvas.drawRRect`, `LinearGradient`, dashed grid lines, and `TextPainter` compact currency labels.
9	**Material 3 Light / Dark Mode & Polish**	✅ Complete	Full Material Design 3 seed color harmonization (`#0284C7`), one-tap Light/Dark mode toggle, and `ExpenseSummaryCard` wrapped in `Dismissible(key: ValueKey(txn.id))`.

#	Required Feature	Status	Implementation Details & Acceptance Level
10	**Auth, GoRouter Guards & Platform Channel**	✅ Complete	`AuthService` + `LoginScreen` (Login/Register, SHA-256 hash, Google OAuth `@vku.udn.vn`, Ephemeral `_obscurePassword`), `GoRouter` `redirect` guard (`/login`), `ShellRoute` 3-tab `NavigationBar` (`/dash`, `/scanner`, `/reports`), `/expense/:id` (`ExpenseDetailScreen`), and `MethodChannel('vn.edu.vku/device_info')`.

3. TECHNICAL ARCHITECTURE & OCR REGEX TUNING

3.1. End-to-End System Pipeline & Clean Architecture

```
[GoRouter Guard (/login)] ──► [Camera Capture / Gallery (/scanner)] ──► [Google ML Kit TextRecognizer (Offline <100ms)]
                                                                                  │
                                                                                  ▼
[Local SQLite (sqflite) + Thumbnail Cache] ◄── [ReceiptReviewScreen (Form Validation)] ◄── [RegexReceiptParser (Dart 3 Records)]
                                                                                  │
                                                                                  ▼
[Riverpod 2 NotifierProvider] ──► [ShellRoute NavigationBar: /dash + /reports + /expense/:id (120Hz CustomPainter)]
```

* **Modular Directory Structure:** lib/core/ (app_theme.dart, app_router.dart) • lib/models/ (user_profile.dart, expense_category.dart, receipt_transaction.dart) • lib/services/ (auth_service.dart, ocr_service.dart, regex_parser_service.dart, storage_service.dart, device_info_channel.dart, mock_receipts_service.dart) • lib/state/ (expense_provider.dart) • lib/widgets/ (custom_pie_chart.dart, custom_bar_chart.dart, camera_overlay_painter.dart, expense_summary_card.dart) • lib/screens/ (login_screen.dart, dashboard_screen.dart, camera_scanner_screen.dart, receipt_review_screen.dart, reports_screen.dart, expense_detail_screen.dart).

3.2. Real-World Vietnamese Receipt Regex Tuning Table

Target Field	Regular Expression Pattern	Edge Cases Handled
Total Keyword Line	`(?:tổng\s*(?:cộng|tiền|thanh\s*toán)?|thanh\s*toán|phải\s*trả|total|amount\s*due)[\s:=-]*([0-9]{1,3}(?:[.,][0-9]{3})*|[0-9]+)`	Matches `TỔNG CỘNG: 87,500 VND`, `THANH TOÁN: 115.000 đ`, `TOTAL: 245000` across noisy thermal prints.
Currency Number Normalization	`\b([0-9]{1,3}(?:[.,][0-9]{3})+)\s*(?:đ|vnd|vnđ)?\b`	Normalizes both dot-separated (`150.000`) and comma-separated (`150,000`) Vietnamese thousands, while rejecting 10-digit phone numbers and Tax IDs (`MST`).
Transaction Date (DD/MM/YYYY)	`\b(0-3]?[0-9])[/\-./\-.\b`	Extracts `28/09/2026`, `12-10-2026`, `12.10.26` and pairs with `([0-2]?[0-9]):([0-5][0-9])` timestamp.

Target Field	Regular Expression Pattern	Edge Cases Handled
Merchant Header & Category	Dictionary lookup (WinMart+, Highlands Coffee, Fahasa, Petrolimex, Thế Giới Di Động) + Top-4 header line heuristic	Filters out generic header words (HÓA ĐƠN, PHIẾU THANH TOÁN, ĐỊA CHỈ, MST) and auto-maps to 6 categories via Dart 3 switch expressions.

4. EMPIRICAL EVIDENCE & SCREENSHOTS (LIGHT & DARK MODES)

* **Evidence 1 — Dashboard in Light & Dark Mode (DashboardScreen):** Displays VKU student header (Nguyễn Thị Thương • 23IT.B219 • 23SE4), Light/Dark theme toggle (🌙 / 🌞), KPI cards (Total Spent & 3,500,000 ₫ Monthly Budget), Category Filter chips, and virtualized ExpenseSummaryCard list with swipe-to-delete (Dismissible).

* **Evidence 2 — Custom Canvas Visualizations (CustomPainter):** Hardware-accelerated Donut Pie Chart (canvas.drawArc) with animated sweep angle and center total counter, paired with 7-day Weekly Bar Chart (canvas.drawRRect) highlighting peak spending days in VKU Sky Blue gradient.

* **Evidence 3 — Live Camera Viewfinder & Framing Overlay (CameraScannerScreen):** Real-time camera feed with 4 glowing corner brackets, animated scanning laser, tap-to-focus (AF) reticle, Flash toggle, and 5 one-tap VKU sample receipt chips (WinMart+, Highlands, Fahasa, Petrolimex, Thế Giới Di Động).

* **Evidence 4 — Interactive OCR Review & Manual Correction (ReceiptReviewScreen):** Shows AI Confidence badge (96% • 24ms), cached receipt thumbnail preview, validated TextFormField inputs for Merchant, Total Amount (VND), Date, Category ChoiceChip selector, and expandable Raw OCR text viewer.

5. TECHNICAL CHALLENGES, RESOLUTIONS & LIMITATIONS

1. **Challenge 1 — Disambiguating Receipt Totals from Phone Numbers & Tax IDs (MST):** Vietnamese receipts contain 10-digit phone numbers (02363667888), tax codes (0104918404), and cash tendered (TIỀN MẶT: 100,000). *Resolution:* Prioritized context-anchored regex lines (TỔNG CỘNG, PHẢI TRẢ) before falling back to max-number heuristics, and filtered out unformatted ≥ 10-digit integers.

2. **Challenge 2 — Zero-Jank 120Hz Custom Chart Animations Without Third-Party Packages:** *Resolution:* Implemented custom shouldRepaint() checks in _DonutChartPainter and _WeeklyBarChartPainter driven by CurvedAnimation(curve: Curves.easeOutCubic), issuing direct drawArc and drawRRect` calls to Flutter's Skia/Impeller canvas.

3. **Current Limitations & Future Work:** Extremely crumpled or faded thermal receipts under low light can degrade OCR character accuracy on Vietnamese diacritics; future iterations will add OpenCV adaptive thresholding pre-processing and multi-page receipt stitching.

Đà Nẵng, Ngày 12 tháng 10 năm 2026

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